



Barker College

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Student Number

2015
TRIAL
HIGHER SCHOOL
CERTIFICATE

Mathematics
Extension 2

AM Friday 31 July

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☐ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ● (B) ~~●~~ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ● (B) ~~●~~ ^{correct} (C) ☐ (D) ☐

Start
Here →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

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Student Number

2015
TRIAL
HIGHER SCHOOL
CERTIFICATE

Mathematics Extension 2

Staff Involved:

AM Friday 31 July

- BHC*
- RMH*
- KJL
- MRB

Number of copies: 50

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided on page 2
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

Pages 2 - 6

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II

Pages 7 - 14

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours 45 minutes for this section

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Section I — Multiple Choice

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 - 10.

1. The eccentricity of the ellipse with equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is

- (A) $\frac{9}{25}$
- (B) $\frac{4}{5}$
- (C) $\frac{3}{5}$
- (D) $\frac{\sqrt{41}}{5}$

2. The equation of degree 10 whose roots are the reciprocals of the equation $x^{10} - 5x^3 + x - 4 = 0$ is

- (A) $4x^{10} - x^9 + 5x^7 - 1 = 0$
- (B) $1 - 5x^7 - x^9 - 4x^{10} = 0$
- (C) $4x^{10} + x^9 + 5x^7 - 1 = 0$
- (D) $4x^{10} - x^9 - 5x^7 + 1 = 0$

3. $P(x) = x^4 + 2x^3 + 9x^2 + 8x + 20$ has a zero $x = 2i - 1$.

What is the value of $P(\overline{2i-1})$?

- (A) $-10 + 12i$
- (B) $-10 - 12i$
- (C) $10 - 12i$
- (D) 0

4. $\int \tan^4 x \, dx$ equals

(A) $\frac{1}{3} \tan^3 x + \tan x + x + c$

(B) $\frac{1}{3} \tan^3 x - \tan x + x + c$

(C) $\frac{1}{3} \tan^3 x - \tan x - x + c$

(D) $\frac{1}{3} \tan^3 x + \tan x - x + c$

5. If w is one of the complex roots of $z^3 = 1$, what is the value of

$(1 - w)(1 - w^2)(1 - w^4)(1 - w^8)?$

(A) 9

(B) 6

(C) 3

(D) 0

6. The vertices of a conic are $(2, 0)$ and $(-2, 0)$ and the foci are $(3, 0)$ and $(-3, 0)$.

What is the equation of this conic?

(A) $\frac{x^2}{4} + \frac{y^2}{5} = 1$

(B) $\frac{x^2}{4} - \frac{y^2}{5} = 1$

(C) $\frac{x^2}{4} - \frac{y^2}{\sqrt{5}} = 1$

(D) $\frac{x^2}{4} + \frac{y^2}{\sqrt{5}} = 1$

7. $\int x e^{-2x} dx$ equals

(A) $\frac{x}{2} e^{-2x} - \frac{1}{2} \int e^{-2x} dx$

(B) $\frac{x}{2} e^{-2x} + \frac{1}{2} \int e^{-2x} dx$

(C) $-\frac{x}{2} e^{-2x} - \frac{1}{2} \int e^{-2x} dx$

(D) $-\frac{x}{2} e^{-2x} + \frac{1}{2} \int e^{-2x} dx$

8. The polynomial $ax^8 + bx^7 + 2$ is divisible by $(x + 1)^2$.

What are the values of a and b ?

(A) $a = -14$ and $b = -16$

(B) $a = 14$ and $b = -16$

(C) $a = 14$ and $b = 16$

(D) $a = -14$ and $b = 16$

9. A unit circle has its centre at the origin O.

The point z_1 moves on the circle and $z_2 = \frac{\sqrt{2} - 3i}{z_1}$.

The Cartesian equation of the locus of z_2 is:

(A) $x^2 - y^2 = 11$

(B) $x^2 + y^2 = 11$

(C) $x^2 + y^2 = \sqrt{11}$

(D) $x^2 - y^2 = \sqrt{11}$

10. All the solutions of the equation $z^4 = (z - 1)^4$ are

(A) $z = \frac{1}{2}, z = \frac{i}{-1 + i}, z = \frac{i}{1 + i}$

(B) $z = \frac{1}{2}, z = \frac{-i}{1 + i}, z = \frac{i}{-1 + i}$

(C) $z = \frac{1}{2}, z = \frac{2i}{-1 + i}, z = \frac{2i}{1 + i}$

(D) $z = \frac{1}{2}, z = \frac{2i}{1 + i}, z = \frac{-2i}{1 + i}$

End of Section I

Section II

90 marks

Attempt Questions 11 - 16.

Allow about 2 hours 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Show relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	[Use a SEPARATE writing booklet]	Marks
(a) (i)	Find the square root of the complex number $-5 + 12i$. Express your answer in the form $a + ib$.	2
(ii)	Hence, or otherwise, solve $2z^2 - (6 + i)z + 5 = 0$ for z . Express your answer in the form $a + ib$.	2
(b)	Prove that $ z_1 - z_2 ^2 = z_1 \bar{z}_1 + z_2 \bar{z}_2 - 2 \operatorname{Re}(z_1 \bar{z}_2)$.	2
(c) (i)	Write down the five roots of $z^5 = -1$.	2
(ii)	Hence show that	
	$z^5 + 1 = (z + 1) \left(z^2 - 2z \cos \frac{\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right)$	3
(d) (i)	If $z = \cos \theta + i \sin \theta$, show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$	1
(ii)	Hence show that	
	$\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$	3

End of Question 11

- (a) Use the substitution $5x - 1 = u^2$ to show that

$$\int x\sqrt{5x-1} \, dx = \frac{2(5x-1)(15x+2)\sqrt{5x-1}}{375} + C \quad 3$$

- (b) Use an appropriate ‘t-result’ substitution to evaluate

$$\int_0^{\frac{\pi}{4}} \frac{d\theta}{2 + \sin 2\theta} \quad 3$$

- (c) (i) Show that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. 1

(ii) Hence find $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} \, dx}{\sqrt{\sin x} + \sqrt{\cos x}}$. 2

- (d) (i) Let $I_n = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^n x \, dx$ and show that $I_n = \frac{-\sqrt{3}}{n2^n} + \left(\frac{n-1}{n}\right) I_{n-2}$ 3

(ii) Hence find $\int_0^{\frac{1}{2}} \frac{x^3 \, dx}{\sqrt{1-x^2}}$ by letting $x = \cos \theta$. 3

End of Question 12

Question 13 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

- (a) The equation $2x^3 - 5x^2 + 8x - 3 = 0$ has roots α , β and γ . **2**
Find α , β and γ , given that one of the roots is $\alpha = 1 - \sqrt{2}i$.
- (b) (i) Prove that if the polynomial $P(x)$ has a root of multiplicity m **2**
at $x = k$ then $P'(x)$ has a root of multiplicity $(m - 1)$ at $x = k$.
- (ii) If $P(x) = 4x^3 + 15x^2 + 12x - 4$ has a double zero, find all the zeros and **3**
factorise $P(x)$ fully over the real numbers.
- (c) The equation $x^4 + 4x^3 - 3x^2 - 4x + 2 = 0$ has roots α , β , γ and δ . **4**
Find the equation with roots α^2 , β^2 , γ^2 and δ^2 .
- (d) One of the roots of the equation $x^3 + ax^2 + bx + c = 0$ is double the sum of the **4**
other two roots. Show that $4a^3 - 18ab + 27c = 0$.

End of Question 13

Question 14 (15 marks)**[Use a SEPARATE writing booklet]****Marks**

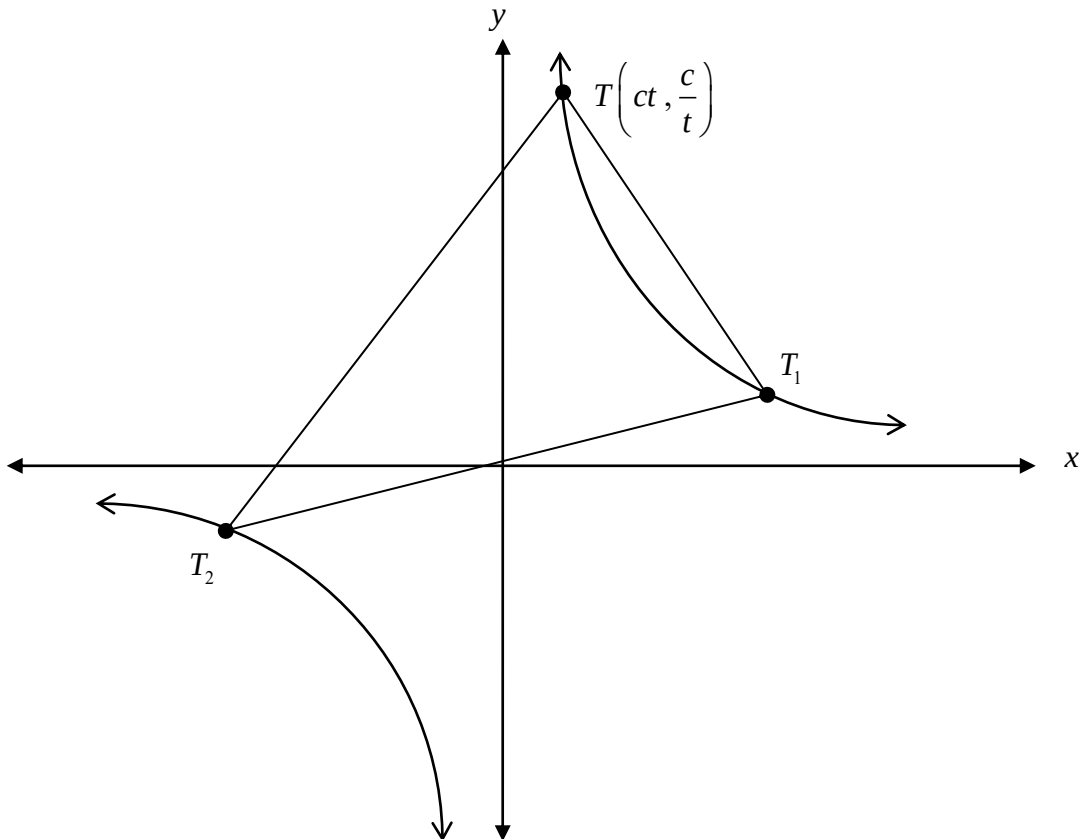
- (a) (i) For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, find the equation of the asymptote with positive gradient. **1**
- (ii) If $P(x_1, y_1)$ is a point on the hyperbola in the first quadrant, show that the equation of the tangent at P is $9xx_1 - 4yy_1 = 36$. **2**
- (iii) If this tangent passes through $(1, 0)$ find the coordinates of P . **1**
- (iv) Let α be the angle between the tangent at P and the asymptote with positive gradient. **3**
Draw a diagram indicating the location of α and show that
$$\alpha = \tan^{-1} \left[\frac{24 - 13\sqrt{3}}{23} \right].$$
- (b) (i) Show that the equation **2**
 $4x^2 - y^2 - 24hx + 2hy - 4a^2 + 35h^2 = 0$, where h and a are positive constants, represents a hyperbola.
- (ii) If the tangent to this hyperbola at the point (p, q) is perpendicular to the straight line **3**
 $y = (e^2 - 1)x$, where e is the eccentricity of the hyperbola,
show that $16p + q = 49h$.

Question 14 continues on page 11

Question 14 (continued)

- (c) The diagram below shows two points, T_1 and T_2 , with parameters t_1 and t_2 respectively, on the rectangular hyperbola $xy = c^2$.

T is a third point with parameter t on the hyperbola such that $\angle T_1 T T_2$ is a right angle.



Show that the gradient of $T_1 T$ is $\frac{-1}{t_1 t}$ and deduce that, since $\angle T_1 T T_2$ is a right-angle, **3**

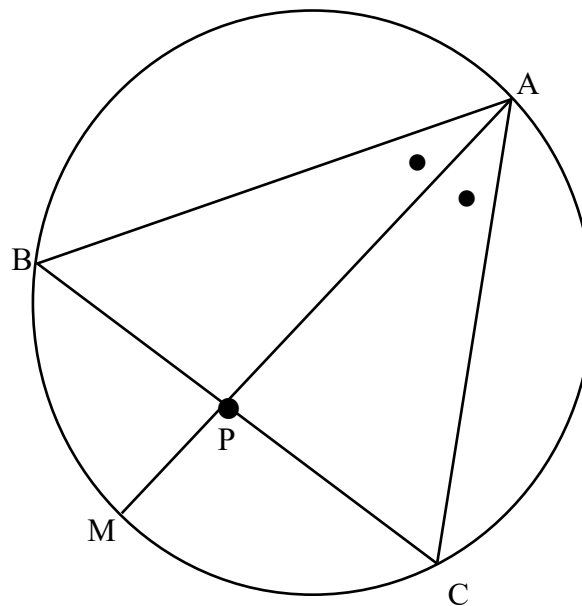
$$t^2 = \frac{-1}{t_1 t_2}.$$

End of Question 14

- (a) Using the method of cylindrical shells, find the volume obtained by revolving the region bounded by the parabola $y = x^2 + 2$, the x -axis, the y -axis and the line $x = 2$, about the line $x = 2$. 4

- (b) The base of a solid is the ellipse $4x^2 + 25y^2 = 100$.
All cross-sections perpendicular to the x -axis are isosceles right triangles (with the smallest side lying in the base).
Find the volume of the solid. 3

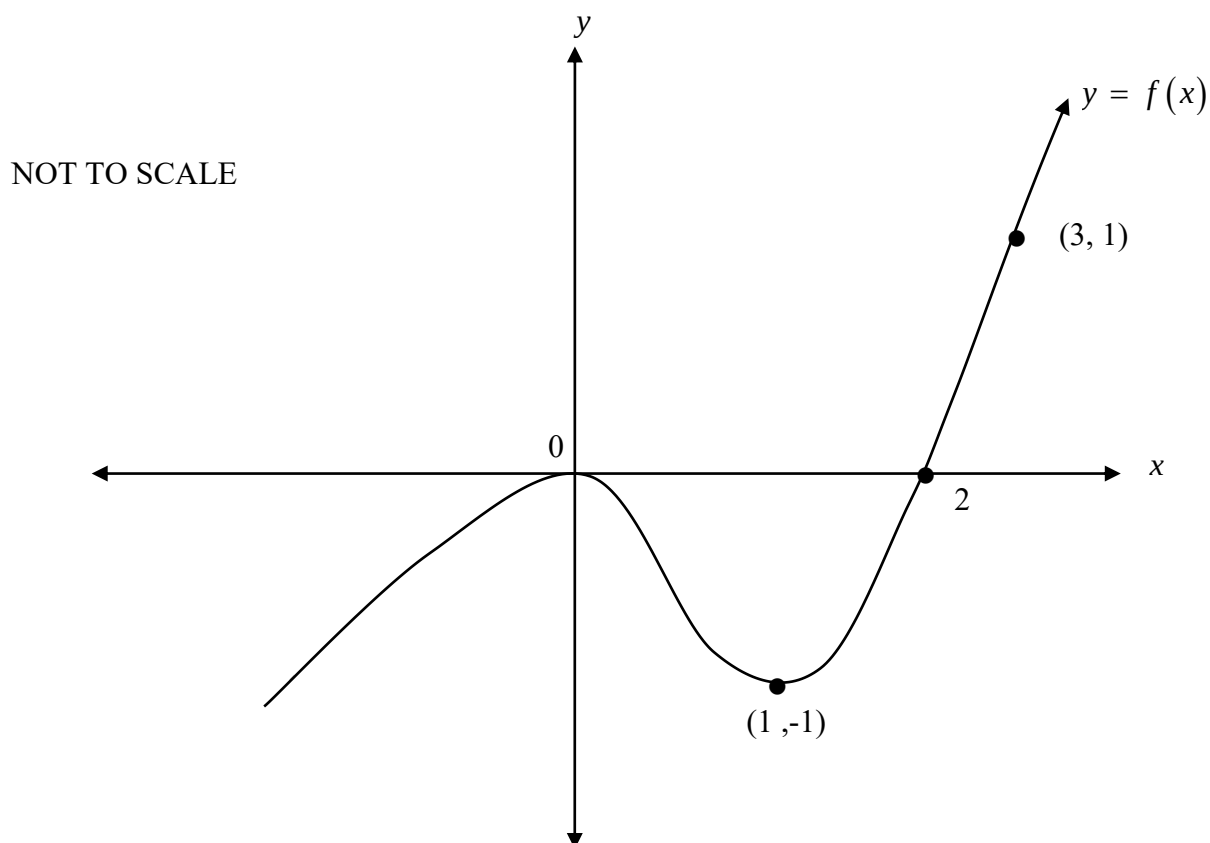
- (c) In the figure below, the bisector AP of $\angle BAC$ is extended to meet the circle in M .



- (i) Prove that $\triangle ABM \parallel \triangle APC$. 2
- (ii) Prove that $BP \times PC = PM \times PA$. 2
- (d) Show by mathematical induction that $35^n + 3 \times 7^n + 2 \times 5^n + 6$ is divisible by 12 for all integers $n \geq 0$. 4

End of Question 15

(a)



The diagram shows the graph of $y = f(x)$.

Draw separate one-third page sketches of the graphs of the following:

(i) $y = |f(|x|)|$ 2

(ii) $y = \ln(f(x))$ 2

Question 16 continues on page 14

Question 16 (continued)

- (b) A spherical planet of mass m_1 and radius R has a rocket launched from its surface with an initial speed of V . The mass of the rocket is m_2 and the distance between the rocket and the centre of the planet is x . The gravitational force, F , acting on the rocket is given by

$$F = \frac{G m_1 m_2}{x^2}$$

where G is the constant of gravitation on that planet.

Assume that there are no other forces acting on the rocket.

- (i) Write down an expression for the acceleration of the rocket in terms of x , taking the positive direction as away from the surface of the planet. 1
- (ii) Find an expression for the velocity v of the rocket in terms of x . 3
- (c) The rise and fall of a tide approximates simple harmonic motion. 4
In a harbour, low tide is at 7am and high tide is at 1:40pm.
The corresponding depths are 20m and 40m.
Find the first time after 7am that a ship which requires $(5\sqrt{3} + 30)$ metres of water is able to enter the harbour.
- (d) When a polynomial $P(x)$ is divided by $x^2 - a^2$, where $a \neq 0$, the remainder is of the form $px + q$ where 3

$$p = \frac{1}{2a} [P(a) - P(-a)] \text{ and } q = \frac{1}{2} [P(a) + P(-a)].$$

(You do not have to prove these results.)

Find the remainder when $P(x) = x^n - a^n$, for n a positive integer, is divided by $x^2 - a^2$.

End of Paper

Multiple Choice

$$Q1_{//} \quad \frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$b^2 = a^2(1 - e^2)$$

$$16 = 25(1 - e^2)$$

$$\frac{16}{25} = 1 - e^2$$

$$e^2 = \frac{9}{25}$$

$$e = \frac{3}{5} \quad (C)$$

Q2_{//} The Required equation is

$$\left(\frac{1}{x}\right)^{10} - 5\left(\frac{1}{x}\right)^3 + \frac{1}{x} - 4 = 0$$

$$1 - 5x^7 + x^9 - 4x^{10} = 0$$

$$4x^{10} - x^9 + 5x^7 - 1 = 0 \quad (A)$$

Q3_{//} $P(x)$ has real coefficients
so complex roots occur in
conjugate pairs. (D)

$$Q4_{//} \quad \int \tan^4 x \, dx$$

$$= \int (\sec^2 x - 1) \cdot \tan^2 x \, dx$$

$$= \int \tan^2 x \cdot \sec^2 x - \tan^2 x \, dx$$

$$= \frac{1}{3} \tan^3 x - \int (\sec^2 x - 1) \, dx$$

$$= \frac{1}{3} \tan^3 x - \tan x + x + C \quad (B)$$

$$Q5_{//} (1-w)(1-w^2)(1-w^4)(1-w^8)$$

$$= (1-w)(1-w^2)(1-w)(1-w^2)$$

$$= [(1-w)(1-w^2)]^2$$

$$= [1 - w - w^2 + w^3]^2$$

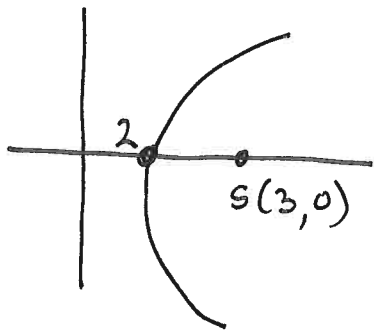
$$= [2 - w - w^2]^2$$

$$\left(\begin{array}{l} \text{but } 1 + w + w^2 = 0 \text{ so} \\ -w - w^2 = 1 \end{array} \right.$$

$$= [2 + 1]^2$$

$$= 9 \quad (A)$$

Q6. //



It is a hyperbola.

$$\text{Hence } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

And when $y=0$, $x=2$

$$\Rightarrow a = 2$$

$$\text{Also } ae = 3$$

$$2e = 3$$

$$e = \frac{3}{2}$$

$$b^2 = a^2(e^2 - 1)$$

$$b^2 = 4\left(\frac{9}{4} - 1\right)$$

$$b^2 = 5$$

$$\text{Hence } \frac{x^2}{4} - \frac{y^2}{5} = 1 \quad \textcircled{B}$$

$$\text{Q7.} // \int x e^{-2x} dx$$

$$= -\frac{1}{2} e^{-2x} \cdot x - \int -\frac{1}{2} e^{-2x} \cdot (1) dx$$

$$= -\frac{x}{2} \cdot e^{-2x} + \frac{1}{2} \int e^{-2x} dx$$

$$= -\frac{x}{2} \cdot e^{-2x} + \frac{1}{2} \left[-\frac{1}{2} e^{-2x} \right]$$

$$= -\frac{x}{2} e^{-2x} + \frac{1}{2} \int e^{-2x} dx \quad \textcircled{B}$$

$$\text{Q8.} // P(x) = ax^8 + bx^7 + 2$$

$$P'(x) = 8ax^7 + 7bx^6$$

$$P(-1) = 0 \Rightarrow a - b + 2 = 0$$

$$P'(-1) = 0 \Rightarrow -8a + 7b = 0$$

$$8a - 8b = -16$$

$$\therefore b = 16$$

$$\therefore a = 14 \quad \textcircled{C}$$

$$\text{Q9.} // |z_1| = 1$$

$$z_1 \times z_2 = \sqrt{2} - 3i$$

$$|z_1 z_2| = |\sqrt{2} - 3i| = \sqrt{11}$$

$$|z_1| \times |z_2| = 11 \quad \therefore |z_2| = \sqrt{11}$$

$$\text{let } z_2 = x + iy$$

$$\text{then } \sqrt{x^2 + y^2} = \sqrt{11}$$

$$\text{ie, } x^2 + y^2 = 11 \quad \textcircled{B}$$

Q10. $z^4 = (z^{-1})^4$

$$\left(\frac{z}{z^{-1}}\right)^4 = 1$$

Hence $\frac{z}{z^{-1}} = 1, -1, i, -i$

If $\frac{z}{z^{-1}} = 1$ then $z = z^{-1}$ (no solutions)

If $\frac{z}{z^{-1}} = -1$ then $z = 1 - z \Rightarrow z = \frac{1}{2}$

If $\frac{z}{z^{-1}} = i$ then $z = iz - i$
 $z(1-i) = -i$
 $z = \frac{-i}{1-i} = \frac{i}{-1+i}$

If $\frac{z}{z^{-1}} = -i$ then $z = -iz + i$
 $z(1+i) = i$
 $z = \frac{i}{1+i}$

Hence (A)

Q11. (a)

(i) Let $(x+iy)^2 = -5+12i$

then $x^2 - y^2 = -5$ and $2xyi = 12i$
 $y = \frac{6}{x}$

Hence $x^2 - \left(\frac{6}{x}\right)^2 = -5$

$$x^4 + 5x^2 - 36 = 0$$

$$(x^2 + 9)(x^2 - 4) = 0$$

$$x = \pm 2$$

When $x = 2$, $y = 3$

When $x = -2$, $y = -3$

Hence the square roots of $-5+12i$ are
 $2+3i$ and $-2-3i$

(ii) $2z^2 - (6+i)z + 5 = 0$

$$z = \frac{6+i \pm \sqrt{(6+i)^2 - 4 \times 2 \times 5}}{4}$$

$$z = \frac{6+i \pm \sqrt{-5+12i}}{4}$$

$$z = \frac{6+i \pm (2+3i)}{4}$$

$$z = 2+i \quad \text{and} \quad 1-\frac{1}{2}i$$

(b) RTP that

$$|z_1 - z_2|^2 = z_1 \bar{z}_1 + z_2 \bar{z}_2 - 2 \operatorname{Re}(z_1 \bar{z}_2)$$

Let $z_1 = a + bi$ and $z_2 = c + di$

$$\begin{aligned} \text{LHS} &= |z_1 - z_2|^2 \\ &= |(a + bi) - (c + di)|^2 \\ &= |(a - c) + i(b - d)|^2 \\ &= \left(\sqrt{(a - c)^2 + (b - d)^2} \right)^2 \\ &= (a - c)^2 + (b - d)^2 \end{aligned}$$

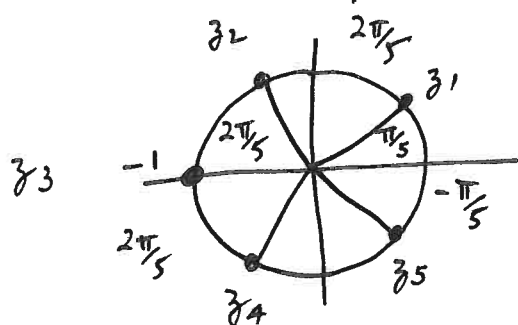
$$\begin{aligned} \text{RHS} &= z_1 \bar{z}_1 + z_2 \bar{z}_2 - 2 \operatorname{Re}(z_1 \bar{z}_2) \\ &= (a + bi)(a - bi) + (c + di)(c - di) - 2 \operatorname{Re}[(a + bi)(c - di)] \\ &= a^2 + b^2 + c^2 + d^2 - 2 \operatorname{Re}[ac + bd + bci - adi] \\ &= a^2 + b^2 + c^2 + d^2 - 2ac - 2bd \\ &= (a - c)^2 + (b - d)^2 \end{aligned}$$

Hence $\text{LHS} = \text{RHS}$ as required.

(c) (i) $z^5 = -1$

$z = -1$ is one solution

The other roots are equally spaced around a unit circle.



The five roots are

$$-1, \operatorname{cis}\left(\frac{\pi}{5}\right), \operatorname{cis}\left(-\frac{\pi}{5}\right), \operatorname{cis}\left(\frac{3\pi}{5}\right), \operatorname{cis}\left(-\frac{3\pi}{5}\right).$$

(ii) Now

$$z^5 + 1 = (z + 1)(z - \operatorname{cis}\left(\frac{\pi}{5}\right))(z - \operatorname{cis}\left(-\frac{\pi}{5}\right)) \times \\ (z - \operatorname{cis}\left(\frac{3\pi}{5}\right))(z - \operatorname{cis}\left(-\frac{3\pi}{5}\right))$$

$$= (z + 1)(z - (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}))(z - (\cos \frac{\pi}{5} - i \sin \frac{\pi}{5})) \times \\ (z - (\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}))(z - (\cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5}))$$

$$= (z + 1)(z^2 - 2z \cos \frac{\pi}{5} + \cos^2 \frac{\pi}{5} + \sin^2 \frac{\pi}{5}) \times \\ (z^2 - 2z \cos \frac{3\pi}{5} + \cos^2 \frac{3\pi}{5} + \sin^2 \frac{3\pi}{5})$$

$$= (z + 1)(z^2 - 2z \cos \frac{\pi}{5} + 1)(z^2 - 2z \cos \frac{3\pi}{5} + 1)$$

as required.

(d) (i) let $z = \cos \theta + i \sin \theta$

then $z^n = (\cos \theta + i \sin \theta)^n$
 $= \cos n\theta + i \sin n\theta$, by de Moivre's Theorem.

Also $z^{-n} = (\cos \theta + i \sin \theta)^{-n}$
 $= \cos(-n\theta) + i \sin(-n\theta)$, by de Moivre.
 $= \cos n\theta - i \sin n\theta$

Hence $z^n - \frac{1}{z^n} = (\cos n\theta + i \sin n\theta) - (\cos n\theta - i \sin n\theta)$
 $= 2i \sin n\theta$ as required.

(ii) RTP that $\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$

Now $z - \frac{1}{z} = 2i \sin \theta$

Hence $(2i \sin \theta)^5 = (z - \frac{1}{z})^5$

$$32i \sin^5 \theta = z^5 - 5z^4\left(\frac{1}{z}\right) + 10z^3\left(\frac{1}{z^2}\right) - 10z^2\left(\frac{1}{z^3}\right) + 5z\left(\frac{1}{z^4}\right) - \left(\frac{1}{z^5}\right)$$

$$= \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$$

$$= 2i \sin 5\theta - 5(2i \sin 3\theta) + 10(2i \sin \theta)$$

$$= 2i [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$$

Hence $\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$

as required.

Q12.

$$(a) \int x \sqrt{5x-1} \, dx$$

$$\text{let } u^2 = 5x-1 \Rightarrow x = \frac{u^2+1}{5}$$

$$2u \frac{du}{dx} = 5$$

$$\frac{2u}{5} du = dx$$

$$I = \int \frac{u^2+1}{5} \cdot \sqrt{u^2} \cdot \frac{2u}{5} du$$

$$= \frac{2}{25} \int u^2(u^2+1) du$$

$$= \frac{2}{25} \int u^4 + u^2 du$$

$$= \frac{2}{25} \left[\frac{u^5}{5} + \frac{u^3}{3} \right] + C$$

$$= \frac{2u^3}{25} \left[\frac{3u^2+5}{15} \right] + C$$

$$= \frac{2}{25} \cdot (5x-1) \cdot \sqrt{5x-1} \cdot \left(\frac{3(5x-1)+5}{15} \right) + C$$

$$= \frac{2(5x-1)\sqrt{5x-1}(15x+2)}{375} + C$$

$$= \frac{2(5x-1)(15x+2)\sqrt{5x-1}}{375} + C, \text{ as required.}$$

$$(b) \int_0^{\pi/4} \frac{d\theta}{2 + \sin 2\theta}$$

$$\text{let } t = \tan \theta$$

$$\frac{dt}{d\theta} = \sec^2 \theta$$

$$\frac{dt}{d\theta} = 1 + t^2$$

$$d\theta = \frac{dt}{1+t^2}$$

$$\text{when } \theta = \frac{\pi}{4}, \quad t = 1$$

$$\text{when } \theta = 0, \quad t = 0$$

$$I = \int_0^1 \frac{\frac{dt}{1+t^2}}{2 + \frac{2t}{1+t^2}}$$

$$= \int_0^1 \frac{dt}{2(1+t^2) + 2t}$$

$$= \int_0^1 \frac{dt}{2(t^2 + t + 1)}$$

$$= \frac{1}{2} \int_0^1 \frac{dt}{(t + \frac{1}{2})^2 + \frac{3}{4}}$$

$$= \frac{1}{2} \int_0^1 \frac{dt}{(\frac{\sqrt{3}}{2})^2 + (t + \frac{1}{2})^2}$$

$$= \frac{1}{2} \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1$$

$$= \left[\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^1$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}(\sqrt{3}) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{1}{\sqrt{3}} \times \frac{\pi}{3} - \frac{1}{\sqrt{3}} \times \frac{\pi}{6}$$

$$= \frac{1}{\sqrt{3}} \times \frac{\pi}{6}$$

$$= \frac{\pi}{6\sqrt{3}} = \frac{\sqrt{3}\pi}{18}$$

(c) (i) RTP that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

$$\text{Let } I = \int_0^a f(a-x) dx$$

$$\text{and let } u = a - x$$

$$\text{then } du = -dx$$

$$\text{so } dx = -du$$

$$\text{When } x = a, u = 0$$

$$\text{When } x = 0, u = a$$

$$\text{Hence } I = \int_a^0 f(u) \cdot (-du)$$

$$= -\int_a^0 f(u) du$$

$$= \int_0^a f(u) du$$

$$= \int_0^a f(x) dx$$

because u is a dummy variable.

$$(ii) \int_0^{\pi/2} \frac{\sqrt{\sin x} \, dx}{\sqrt{\sin x} + \sqrt{\cos x}}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - x)} \, dx}{\sqrt{\sin(\pi/2 - x)} + \sqrt{\cos(\pi/2 - x)}}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos x} \, dx}{\sqrt{\cos x} + \sqrt{\sin x}}$$

$$\text{Hence } 2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx$$

$$= \int_0^{\pi/2} 1 \, dx$$

$$= \frac{\pi}{2}$$

$$\text{Hence } I = \frac{\pi}{4}$$

$$(d) (i) I_n = \int_{\pi/3}^{\pi/2} \cos^n x \, dx$$

$$= \int_{\pi/3}^{\pi/2} \cos x \cdot \cos^{n-1} x \, dx$$

$$= \left[\sin x \cdot \cos^{n-1} x \right]_{\pi/3}^{\pi/2}$$

$$- \int_{\pi/3}^{\pi/2} \sin x \cdot (n-1) \cos^{n-2} x \cdot (-\sin x) \, dx$$

$$= -\sin \frac{\pi}{3} \times \left(\cos \frac{\pi}{3} \right)^{n-1} + \int_{\pi/3}^{\pi/2} \sin^2 x \cdot (n-1) \cos^{n-2} x \, dx$$

$$= -\frac{\sqrt{3}}{2} \cdot \left(\frac{1}{2} \right)^{n-1} + (n-1) \int_{\pi/3}^{\pi/2} (1 - \cos^2 x) \cdot \cos^{n-2} x \, dx$$

$$= -\frac{\sqrt{3}}{2^n} + (n-1) \int_{\pi/3}^{\pi/2} \cos^{n-2} x \, dx - (n-1) \int_{\pi/3}^{\pi/2} \cos^n x \, dx$$

$$= -\frac{\sqrt{3}}{2^n} + (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore (n-1) I_n + I_n = -\frac{\sqrt{3}}{2^n} + (n-1) I_{n-2}$$

$$n I_n = -\frac{\sqrt{3}}{2^n} + (n-1) I_{n-2}$$

$$\therefore I_n = \frac{-\sqrt{3}}{n 2^n} + \left(\frac{n-1}{n} \right) I_{n-2}$$

$$(ii) \int_0^{\frac{1}{2}} \frac{x^3 dx}{\sqrt{1-x^2}}$$

Let $x = \cos \theta$

$$dx = -\sin \theta d\theta$$

When $x = \frac{1}{2}$, $\theta = \frac{\pi}{3}$

When $x = 0$, $\theta = \frac{\pi}{2}$

Hence
$$I = \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{\cos^3 \theta \cdot (-\sin \theta d\theta)}{\sqrt{1-\cos^2 \theta}}$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^3 \theta d\theta = I_3$$

And
$$I_3 = \frac{-\sqrt{3}}{3 \times 2^3} + \frac{2}{3} I_1 \quad (\text{from part (i)})$$

And
$$I_1 = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos x dx$$

$$= \left[\sin x \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= 1 - \frac{\sqrt{3}}{2}$$

Hence
$$I_3 = \frac{-\sqrt{3}}{24} + \frac{2}{3} \left(1 - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{-\sqrt{3}}{24} + \frac{16}{24} - \frac{8\sqrt{3}}{24} = \frac{16 - 9\sqrt{3}}{24}$$

QB, (a)

$$2x^3 - 5x^2 + 8x - 3 = 0 \text{ has } \alpha = 1 - \sqrt{2}i$$

Hence $\beta = 1 + \sqrt{2}i$ since complex roots occur in conjugate pairs when all coefficients are real.

$$\text{But } \Sigma(\alpha) = -\frac{b}{a} = \frac{5}{2}$$

$$\text{Hence } (1 - \sqrt{2}i) + (1 + \sqrt{2}i) + \gamma = \frac{5}{2}$$
$$\gamma = \frac{1}{2}$$

$$(b) (i) \text{ Let } P(x) = (x-k)^m \cdot Q(x)$$

$$\text{then } P'(x) = (x-k)^m \cdot Q'(x) + Q(x) \times m(x-k)^{m-1}$$
$$= (x-k)^{m-1} [(x-k) \cdot Q'(x) + mQ(x)]$$

$$\therefore P'(k) = P(k) = 0$$

$\therefore P'(x)$ has a root of multiplicity $(m-1)$

$$(b) (ii) \quad P(x) = 4x^3 + 15x^2 + 12x - 4$$

$$P'(x) = 12x^2 + 30x + 12$$

$$= 6(2x^2 + 5x + 2)$$

$$= 6(2x+1)(x+2)$$

$$\therefore P'(x) = 0 \quad \text{when } x = -\frac{1}{2} \quad \text{and } x = -2$$

$$\text{But } P(-\frac{1}{2}) = -\frac{4}{8} + \frac{15}{4} - 6 - 4 \neq 0$$

$$\begin{aligned} \text{and } P(-2) &= -4 \times 8 + 60 - 24 - 4 \\ &= -32 + 60 - 28 \\ &= 0 \end{aligned}$$

Hence $x = -2$ is the double zero.

$$\text{Hence } P(x) = (x+2)^2(ax+b)$$

but $a = 4$ and $b = -1$ by inspection.

$$\text{Hence } P(x) = (x+2)^2(4x-1)$$

and the zeros are $-2, -2, \frac{1}{4}$.

(c) The required equation is

$$(\sqrt{x})^4 + 4(\sqrt{x})^3 - 3(\sqrt{x})^2 - 4(\sqrt{x}) + 2 = 0$$

$$x^2 + 4x\sqrt{x} - 3x - 4\sqrt{x} + 2 = 0$$

$$x^2 - 3x + 2 = 4\sqrt{x}(1-x)$$

$$(x^2 - 3x + 2)(x^2 - 3x + 2) = 16x(1-x)^2$$

$$x^4 - 3x^3 + 2x^2 - 3x^3 + 9x^2 - 6x + 2x^2 - 6x + 4 = 16x - 32x^2 + 16x^3$$

$$x^4 - 6x^3 + 13x^2 - 12x + 4 = 16x - 32x^2 + 16x^3$$

$$\text{ie, } x^4 - 22x^3 + 45x^2 - 28x + 4 = 0$$

(d) Let the roots of $x^3 + ax^2 + bx + c = 0$ be α , β and γ where $\alpha = 2(\beta + \gamma)$.

Now the sum of the roots is

$$\alpha + \beta + \gamma = -a$$

$$\text{ie, } \alpha + \frac{\alpha}{2} = -a$$

$$\frac{3\alpha}{2} = -a$$

$$\alpha = \frac{-2a}{3}$$

And the sum of the product of the roots 2 at a time is

$$\alpha\beta + \beta\gamma + \alpha\gamma = b$$

$$2(\beta + \gamma) \cdot \beta + \beta\gamma + 2(\beta + \gamma) \cdot \gamma = b$$

$$(\beta + \gamma)(2\beta + 2\gamma) + \beta\gamma = b$$

$$2(\beta + \gamma)(\beta + \gamma) + \beta\gamma = b$$

$$2\left(\frac{\alpha}{2}\right)\left(\frac{\alpha}{2}\right) + \beta\gamma = b$$

$$\beta\gamma = b - \frac{\alpha^2}{2}$$

$$\beta\gamma = b - \frac{\left(\frac{-2a}{3}\right)^2}{2}$$

$$\beta\gamma = b - \frac{4a^2}{18} \quad \text{--- (1)}$$

And the product of the roots 3 at a time is

$$\alpha\beta\gamma = -c$$

$$\beta\gamma = \frac{-c}{\alpha} = \frac{-c}{-2a/3} = \frac{3c}{2a} \quad \text{--- (2)}$$

Equating (1) and (2) gives

$$\frac{3c}{2a} = b - \frac{4a^2}{18}$$

$$3c = 2ab - \frac{8a^3}{18}$$

$$54c = 36ab - 8a^3$$

$$4a^3 - 18ab + 27c = 0$$

ie,

Q14. //

(a) (i) $y = \frac{3}{2}x$

(ii) $\frac{x^2}{4} - \frac{y^2}{9} = 1$

$$\frac{2x}{4} - \frac{2y}{9} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{18x}{8y} = \frac{9x}{4y}$$

Hence the gradient of the tangent at (x_1, y_1) is $\frac{9x_1}{4y_1}$

Hence the equation of the tangent is

$$y - y_1 = \frac{9x_1}{4y_1} (x - x_1)$$

$$4yy_1 - 4(y_1)^2 = 9xx_1 - 9(x_1)^2$$

$$9(x_1)^2 - 4(y_1)^2 = 9xx_1 - 4yy_1$$

But from $\frac{x^2}{4} - \frac{y^2}{9} = 1$ $9x^2 - 4y^2 = 36$

So $9(x_1)^2 - 4(y_1)^2 = 36$

Hence the equation of the tangent is

$$9xx_1 - 4yy_1 = 36$$

$$(iii) \quad 9xx_1 - 4yy_1 = 36$$

passes through $(1, 0)$ so

$$9x_1 = 36$$

$$x_1 = 4$$

$$\text{And } 9(x_1)^2 - 4(y_1)^2 = 36$$

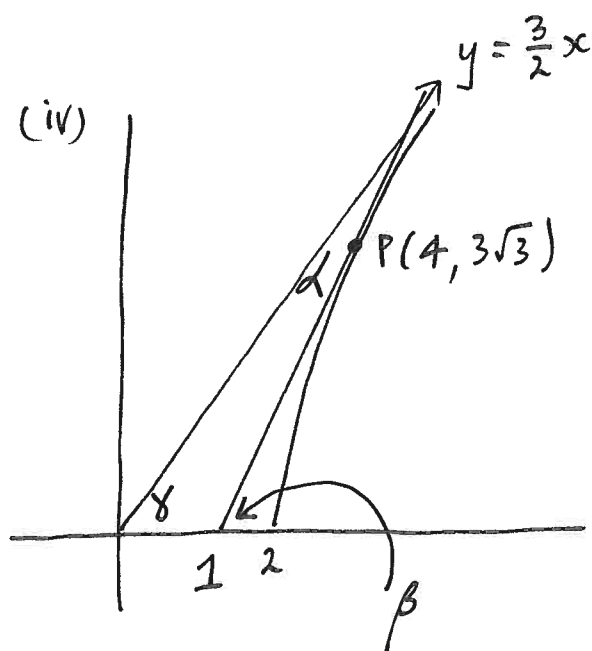
$$9 \times 4^2 - 4(y_1)^2 = 36$$

$$108 = 4(y_1)^2$$

$$y_1 = \sqrt{27}$$

$$y_1 = 3\sqrt{3}$$

$$\text{Hence } (x_1, y_1) = (4, 3\sqrt{3})$$



$$\beta = \alpha + \gamma \quad (\text{exterior angle})$$

$$\alpha = \beta - \gamma$$

$$\begin{aligned} \tan \alpha &= \tan(\beta - \gamma) \\ &= \frac{\tan \beta - \tan \gamma}{1 + \tan \beta \cdot \tan \gamma} \end{aligned}$$

$$\text{But } \tan \beta = \frac{3\sqrt{3}}{4-1} = \sqrt{3} \quad \text{and } \tan \gamma = \frac{3}{2}$$

$$\text{Hence } \tan \alpha = \frac{\sqrt{3} - \frac{3}{2}}{1 + \sqrt{3} \times \frac{3}{2}} = \frac{2\sqrt{3} - 3}{2 + 3\sqrt{3}} \times \frac{2 - 3\sqrt{3}}{2 - 3\sqrt{3}} = \frac{-24 + 13\sqrt{3}}{-23}$$

$$\therefore \alpha = \tan^{-1} \left[\frac{24 - 13\sqrt{3}}{23} \right]$$

(b) (i)

$$4x^2 - y^2 - 24hx + 2hy - 4a^2 + 35h^2 = 0$$

$$4x^2 - 24hx - y^2 + 2hy = 4a^2 - 35h^2$$

$$(2x - 6h)^2 - (y - h)^2 - 36h^2 + h^2 = 4a^2 - 35h^2$$

$$(2x - 6h)^2 - (y - h)^2 = 4a^2$$

$$4(x - 3h)^2 - (y - h)^2 = 4a^2$$

$$\frac{(x - 3h)^2}{a^2} - \frac{(y - h)^2}{4a^2} = 1 \quad \text{--- (1)}$$

(1) is of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Hence the equation represents a hyperbola.

(ii) Differentiating (1) we get

$$\frac{2(x - 3h)}{a^2} - \frac{2(y - h)}{4a^2} \times \frac{dy}{dx} = 0$$

$$\text{So } \frac{dy}{dx} = \frac{2 \times 4a^2 (x - 3h)}{2 \times a^2 (y - h)}$$

$$\frac{dy}{dx} = \frac{4(x - 3h)}{(y - h)}$$

Hence the gradient of the tangent at (p, q) is

$$\frac{4(p-3h)}{q-h}.$$

Now the gradient of the line perpendicular to $y = (e^2-1)x$ is $\frac{-1}{e^2-1}$

$$\text{Hence } \frac{-1}{e^2-1} = \frac{4(p-3h)}{q-h} \dots\dots \textcircled{1}$$

$$\text{But } b^2 = a^2(e^2-1)$$

$$\text{so } e^2-1 = \frac{b^2}{a^2}$$

$$\text{But } b^2 = 4a^2 \text{ so } e^2-1 = 4.$$

Substituting into $\textcircled{1}$ we get

$$\frac{-1}{4} = \frac{4(p-3h)}{q-h}$$

$$h-q = 16p - 48h$$

$$\therefore 16p + q = 49h, \text{ as required.}$$

(c) Now $T_1 = (ct_1, \frac{c}{t_1})$

and $T = (ct, \frac{c}{t})$

Hence m of $T_1 T$ is $\frac{\frac{c}{t} - \frac{c}{t_1}}{ct - ct_1} = \frac{\frac{1}{t} - \frac{1}{t_1}}{t - t_1}$

$$= \frac{\frac{t_1 - t}{t_1 t}}{t - t_1} = \frac{-\frac{(t - t_1)}{t_1 t}}{t - t_1} = \frac{-1}{t_1 t}$$

Similarly m of $T_2 T$ is $\frac{-1}{t_2 t}$.

Since $\angle T_1 T T_2$ is 90° then

$$(m \text{ of } T_1 T) \times (m \text{ of } T_2 T) = -1$$

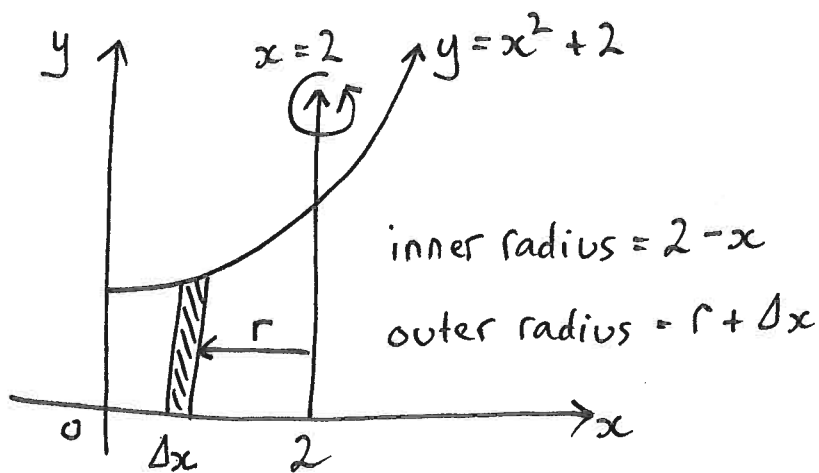
$$\frac{-1}{t_1 t} \times \frac{-1}{t_2 t} = -1$$

$$\therefore \frac{1}{t_1 t_2 t^2} = -1$$

$$\therefore \frac{1}{t_1 t_2} = -t^2$$

$$\therefore t^2 = \frac{-1}{t_1 t_2}, \text{ as required.}$$

Q15. (a)



Rotation of the shaded strip about the line $x = 2$ produces a cylindrical shell of volume ΔV where

$$\Delta V = \pi [(r + \Delta x)^2 - r^2] \times y$$

$$= \pi [(2 - x + \Delta x) \Delta x] \times (x^2 + 2)$$

$$= \pi [2 - x] \Delta x (x^2 + 2) \quad \text{because } (\Delta x)^2 \text{ is very small}$$

$$= \pi [2(2 - x)(x^2 + 2) \Delta x]$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \Delta V$$

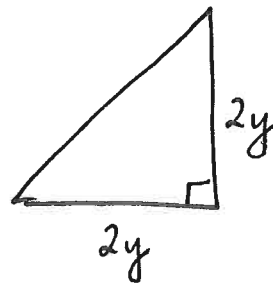
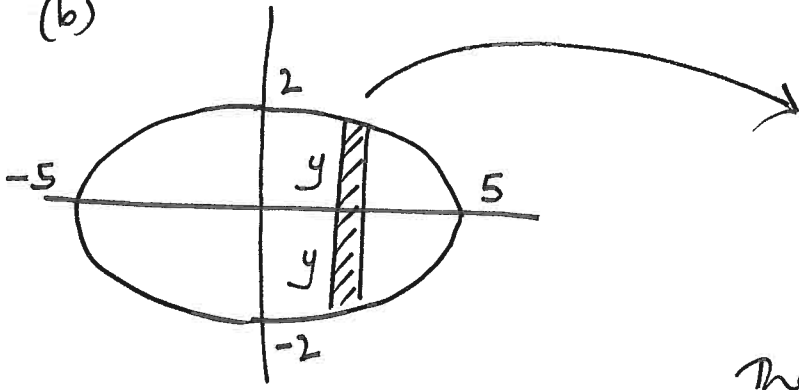
$$= 2\pi \int_0^2 (2 - x)(x^2 + 2) dx$$

$$= 2\pi \int_0^2 (2x^2 + 4 - x^3 - 2x) dx$$

$$= 2\pi \left[\frac{2x^3}{3} + 4x - \frac{x^4}{4} - x^2 \right]_0^2$$

$$= 2\pi \left(\frac{16}{3} + 8 - 4 - 4 \right) = \frac{32\pi}{3} \text{ units}^3$$

(b)



The volume of a slice is ΔV .

$$\begin{aligned}\Delta V &= \frac{1}{2} \times 2y \times 2y \times \Delta x \\ &= 2y^2 \Delta x\end{aligned}$$

Now $y^2 = \frac{100 - 4x^2}{25} \quad \therefore \Delta V = 2 \left(\frac{100 - 4x^2}{25} \right) \Delta x$

The required volume is V where

$$V = \lim_{\Delta x \rightarrow 0} \Delta V$$

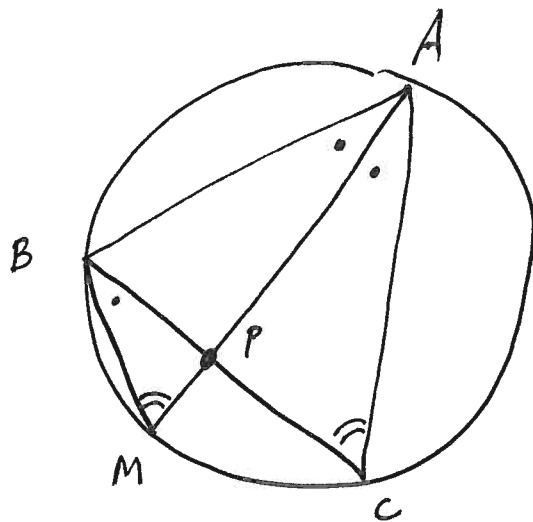
$$= \int_{-5}^5 \frac{2}{25} (100 - 4x^2) dx$$

$$= \int_{-5}^5 \frac{8}{25} (25 - x^2) dx$$

$$= \frac{16}{25} \int_0^5 (25 - x^2) dx$$

$$= \frac{16}{25} \left[25x - \frac{x^3}{3} \right]_0^5 = \frac{160}{3} \text{ units}^3$$

(c)



(i) Join B to M.

In $\triangle ABM$ and $\triangle PCA$,

$$\angle BAP = \angle PAC \text{ (given)}$$

$$\angle BMA = \angle ACP$$

(equal angles at the circumference standing on the same arc AB)

Hence $\triangle ABM \parallel \triangle APC$ (equiangular)

(ii) Now $\angle MBC = \angle CAP$

(equal angles standing on arc MC)

$\therefore \triangle BMP \parallel \triangle APC$ (equiangular)

$$\therefore \frac{BP}{PA} = \frac{PM}{PC} \text{ (ratios of corresponding sides of similar triangles)}$$

$$\therefore BP \times PC = PM \times PA, \text{ as required.}$$

(d) STEP 1: When $n=0$,

$35^n + 3 \times 7^n + 2 \times 5^n + 6$ is equal to

$$1 + 3 \times 1 + 2 \times 1 + 6 = 12, \text{ which is divisible by } 12.$$

Hence the statement is true when $n=0$.

STEP 2: We will assume that

$$35^k + 3 \times 7^k + 2 \times 5^k + 6 = 12M \quad (M \text{ an integer}).$$

We will now try to prove that

$35^{k+1} + 3 \times 7^{k+1} + 2 \times 5^{k+1} + 6$ is divisible by 12.

$$\text{Now } 35^{k+1} + 3 \times 7^{k+1} + 2 \times 5^{k+1} + 6$$

$$= 35 \cdot 35^k + 3 \cdot 7^{k+1} + 2 \cdot 5^{k+1} + 6$$

$$\left(\begin{array}{l} \text{but } 35^k = 12M - 6 - 2 \cdot 5^k - 3 \cdot 7^k \text{ (by the assumption)} \\ \downarrow \end{array} \right)$$

$$= 35(12M - 6 - 2 \cdot 5^k - 3 \cdot 7^k) + 3 \cdot 7^{k+1} + 2 \cdot 5^{k+1} + 6$$

$$= 420M - 210 - 70 \cdot 5^k - 105 \cdot 7^k + 3 \cdot 7^{k+1} + 2 \cdot 5^{k+1} + 6$$

$$= 420M - 210 - 70 \cdot 5^k - 105 \cdot 7^k + 21 \cdot 7^k + 10 \cdot 5^k + 6$$

$$= 420M - 210 - 60 \cdot 5^k - 84 \cdot 7^k + 6$$

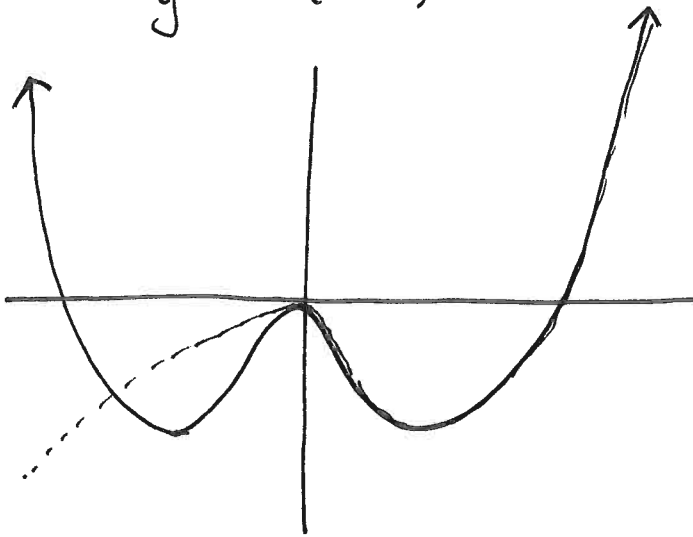
$$= 420M - 204 - 60 \cdot 5^k - 84 \cdot 7^k$$

$$= 12[35M - 17 - 5^{k+1} - 7^{k+1}] \text{ which is divisible by } 12.$$

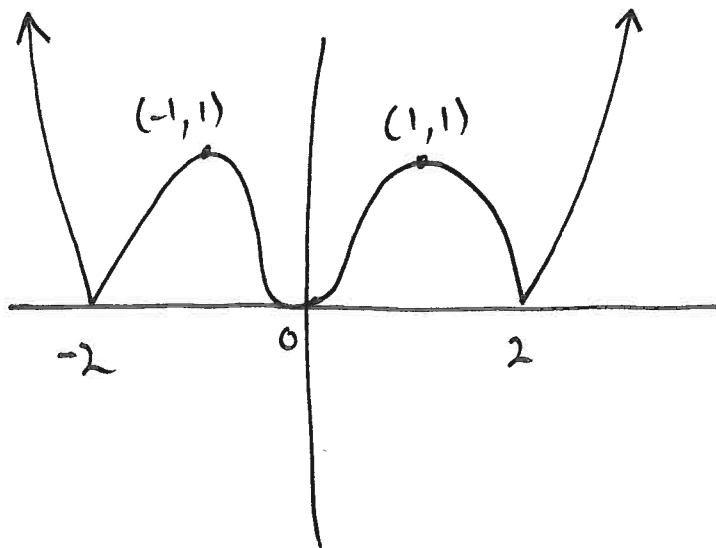
Hence the statement is true by mathematical induction.

Q16. (a)

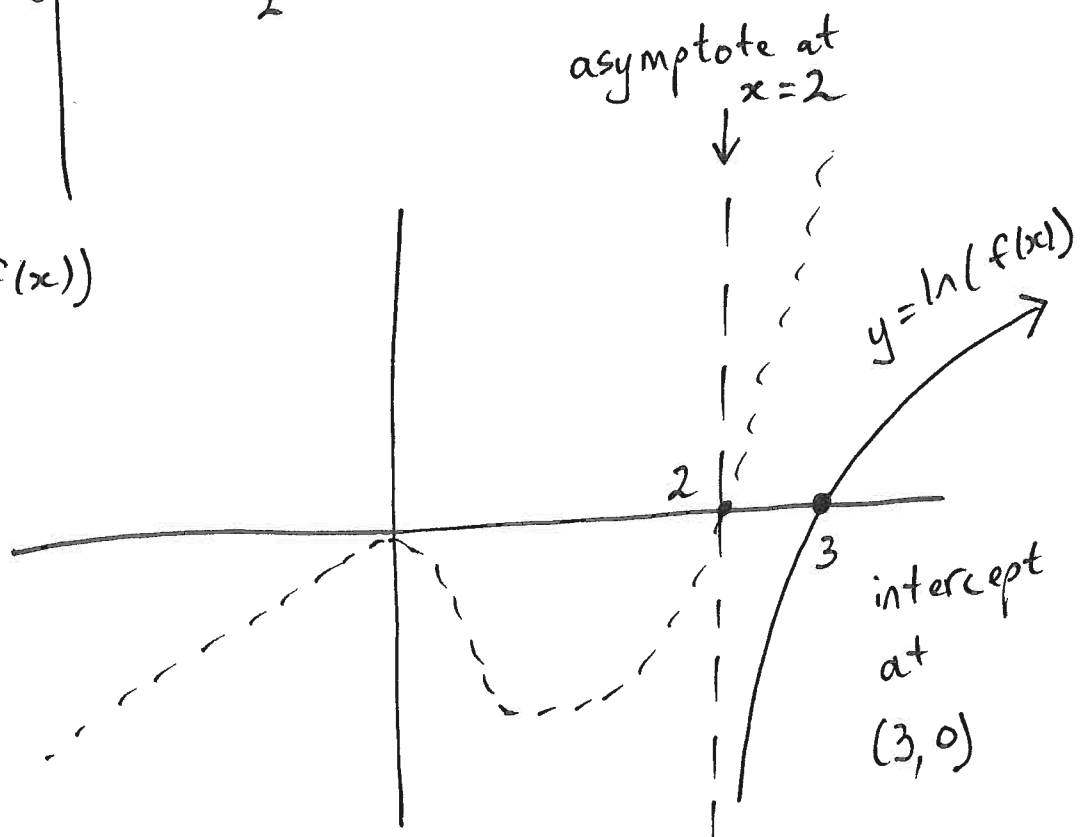
(i) $y = f(|x|)$



$y = |f(|x|)|$



(ii) $y = \ln(f(x))$



$$(b) \quad (i) \quad F = \frac{G m_1 m_2}{x^2}$$

But $F = m_2 \times \text{acceleration}$

$$\therefore m_2 \ddot{x} = \frac{-G m_1 m_2}{x^2}$$

$$\text{Hence } \ddot{x} = \frac{-G m_1}{x^2}$$

$$(ii) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{-G m_1}{x^2}$$

$$\frac{1}{2} v^2 = -G m_1 \int \frac{dx}{x^2}$$

$$\frac{1}{2} v^2 = -G m_1 \left(-\frac{1}{x} \right) + C$$

But when $x = R$, $v = V$ so

$$\frac{1}{2} V^2 = \frac{G m_1}{R} + C$$

$$\therefore C = \frac{1}{2} V^2 - \frac{G m_1}{R}$$

$$\therefore \frac{1}{2} v^2 = \frac{G m_1}{x} + \frac{1}{2} V^2 - \frac{G m_1}{R}$$

$$v^2 = \frac{2G m_1}{x} + V^2 - \frac{2G m_1}{R}$$

$$\therefore v = \sqrt{V^2 + \frac{2G m_1}{x} - \frac{2G m_1}{R}}$$

(c) The time between low and high tide is 6h 40 min.

$$\begin{aligned}\text{So the period of tide} &= 13\text{h } 20\text{mins} \\ &= \frac{40}{3} \text{ hours}\end{aligned}$$

$$\text{But } T = \frac{2\pi}{n} = \frac{40}{3} \Rightarrow n = \frac{3\pi}{20}$$

The centre of motion is at 30m and the amplitude is 10m.

$$\begin{aligned}\text{Now } x &= b + a \cos(nt + \alpha) \\ &= 30 + 10 \cos(nt + \alpha)\end{aligned}$$

Let $t=0$ when $x=20$ (ie, at 7am) then

$$20 = 30 + 10 \cos \alpha$$

$$\alpha = -\pi$$

$$\text{Hence } x = 30 + 10 \cos\left(\frac{3\pi}{20}t - \pi\right)$$

$$\text{Solve } 5\sqrt{3} + 30 = 30 + 10 \cos\left(\frac{3\pi}{20}t - \pi\right)$$

$$\frac{\sqrt{3}}{2} = \cos\left(\frac{3\pi t}{20} - \pi\right)$$

$$-\frac{\pi}{6} = \frac{3\pi t}{20} - \pi$$

$$t = \frac{50}{9} \text{ hours}$$

$\therefore$ Ship can enter at 12:33 pm

(d) Now $P(x) = x^n - a^n$

and $x^n - a^n = (x^2 - a^2) \cdot Q(x) + px + q$.

When n is even, $P(a) = 0$

$$P(-a) = 0$$

Hence $p = 0$ and $q = 0$.

Hence there is no remainder when n is even.

when n is odd, $P(a) = 0$

$$P(-a) = -a^n - a^n = -2a^n$$

$$\therefore px + q = \frac{1}{2a} [0 - (-2a^n)]x + \frac{1}{2} [0 - 2a^n]$$

$$= a^{n-1}x - a^n$$

when n is odd, the remainder is $a^{n-1}x - a^n$.